

Distributed Denial-of-Service

How Computers Learn to Stop Cyber Attacks

A plain-English guide to: A Comparative Hybrid Framework
for Real-Time DDoS Detection Using Machine Learning

The Problem

The Solution

3-Layer AI System

Real Impact

Future Projects

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1. The Problem

What Is a DDoS Attack and Why Is It So Hard to Stop?

1.1 A Real-World Analogy

Imagine you own a small shop with one door. Your shop can serve 10 customers at a time. Now imagine 10,000 strangers show up, crowd every inch of space, block the entrance, and refuse to leave — even though none of them intend to buy anything. Real customers cannot get in. Your business shuts down. That is exactly what a DDoS attack does to a website or online service.

DDoS = Distributed Denial-of-Service

A DDoS attack floods a server with so much fake internet traffic that real users cannot reach it. 'Distributed' means the attack comes from thousands of different computers simultaneously — making it impossible to simply block one source.

Real example: In 2016, an attack on the company Dyn took down Twitter, Netflix, Reddit, and Amazon for hours. The attack used hacked home routers, CCTV cameras, and baby monitors as weapons — costing millions of dollars in lost revenue.

1.2 Why Is This Such a Hard Problem?

Challenge	Why It Matters
Scale Problem	A server receives millions of packets per second. An attack adds tens of millions of fake ones. Finding attack packets is like identifying the wrong grains of sand on a beach — while a new truck of sand arrives every second.
Speed Problem	Every second of downtime costs money. A large e-commerce site loses roughly \$80,000 per minute. A detection system that takes even 30 seconds to respond is already too slow to prevent damage.
Disguise Problem	Modern attackers craft fake traffic to look exactly like real user traffic — same timing, same packet sizes, same protocols. Old pattern-matching systems are fooled completely.
Flow Completion Problem	Traditional tools inspect network flows only after they finish — which takes 30 to 120 seconds. By then the server is already overwhelmed. It is like reviewing CCTV footage after the robbery is already over.

Table 1: Four core challenges that make real-time DDoS detection difficult.

1.3 What Was Being Done Before — and Why It Failed

OLD METHOD 1 — Signature-Based Detection

How it works: Security engineers write rules like 'if a packet looks like THIS, block it.' Rules are based on known attack patterns.

Why it fails: The moment an attacker makes a small change — different timing, different packet size — the rule no longer matches. It is like having a list of known criminals' faces, but the criminal shaved his beard.

OLD METHOD 2 — Flow-Based Detection

How it works: The system waits for a complete network conversation (called a flow) to finish, then analyses the entire conversation.

Why it fails: A conversation takes 30 to 120 seconds to complete. The attack is happening during that entire time. Detecting it afterwards is useless for preventing the damage.

Neither approach could detect attacks in real time against patterns never seen before. That gap is exactly what this research fills.

2. The Solution

Teaching Computers to Detect Attacks in Real Time

2.1 The Core Idea: Watch Traffic Live

Instead of waiting for a network conversation to finish, this system watches traffic as it arrives — like a security guard at the door watching every person who enters, rather than reviewing photos after the event. Every second, the system computes five numbers from the network traffic and asks: is this normal or an attack?

The Five Numbers Computed Every Second

1. Packet Rate — How many packets arrived per second? Normal: ~1,000/sec. Attack: 500,000+/sec.
2. Mean Packet Size — What is the average size of each packet? Normal: varies widely. Attack: very uniform.
3. Size Variance — How different are the packet sizes from each other? Normal: high variety. Attack: nearly identical.
4. Inter-Arrival Time — How much time passes between each packet? Normal: irregular gaps. Attack: machine-precise, extremely short.
5. Byte Rate — How many total bytes arrived per second? Normal: moderate and variable. Attack: extremely high and steady.

These five numbers come from a **sliding one-second window** that moves forward every half-second — a 50% overlap ensures no attack burst falls between two windows and goes undetected.

2.2 The Three-Layer Detection System

Once the five numbers are computed, the traffic passes through three layers — from the fastest and cheapest check to the most thorough. Think of it as three security checkpoints at an airport.

LAYER 1 — The Quick Check (The Bouncer at the Door)

Speed: under 0.01 milliseconds | Method: Simple rules — no AI required

What it does: Two fast rules are checked first. If more than 50,000 packets arrive per second, OR if packet size variance drops below 10 bytes squared, the traffic is immediately flagged as an attack. No machine learning model is invoked.

Real-world analogy: A concert venue bouncer has one rule: if more than 500 people try to enter in one minute, close the door immediately. No ID check needed — the sheer number is the red flag. The rule is crude but catches the most obvious threats at zero cost.

Why this layer exists: 90% of DDoS attacks are simple volumetric floods. This rule catches them in under 0.01 milliseconds, before any AI model is even started. It filters the vast majority of attack traffic at negligible processing cost.

LAYER 2 — The Suspicious Behaviour Detector (The Plainclothes Store Detective)

Speed: 0.18 milliseconds | Method: Isolation Forest — AI trained on normal traffic only

What it does: Traffic that passed Layer 1 goes to an AI model trained exclusively on normal, benign traffic. It has never been shown an attack. It computes an anomaly score from 0 to 1. If the score exceeds 0.6 — meaning the traffic looks too unusual compared to anything normal — it is flagged.

Real-world analogy: A plainclothes store detective does not know every shoplifting technique. They know how normal shoppers behave. If someone is walking suspiciously slowly, touching every item but buying nothing, never checking price tags — that person gets followed. The detector knows 'normal behaviour'; anything far from normal triggers suspicion.

How it works: The algorithm randomly partitions the traffic data space. Anomalous points get isolated very quickly — in very few splits — because they are unusual and separated from the dense cluster of normal points. Normal points take many splits to isolate. Score near 1.0 = attack; near 0.5 = normal.

Why this layer exists: New types of attacks that no one has ever seen before will not match any rule. But they will still look statistically unusual compared to normal traffic. This layer catches zero-day (brand-new) attacks that Layer 1 cannot.

LAYER 3 — The Expert Investigator (The Forensic Analyst)

Speed: 0.41 milliseconds | Method: Random Forest — AI trained on labelled attack and normal traffic

What it does: Traffic that passed both earlier layers reaches this final, most thorough stage. The Random Forest AI uses all 68 features extracted from the full packet flow to make a precise binary decision: Attack or Normal.

Real-world analogy: A forensic specialist examines everything — footprints, CCTV footage, fingerprints, witness statements, the full timeline. They use ALL available evidence. This is the most expensive layer, but also the most accurate — and it is called in only when the first two layers were inconclusive.

How it works: Random Forest builds 100 independent decision trees, each trained on a different random sample of data. Each tree votes: Attack or Normal. The majority vote wins. Using 100 independent opinions makes the system extremely reliable — one flawed tree cannot sway the group.

Why this layer exists: It is the most computationally expensive layer. Placing it last means most traffic — already resolved by Layers 1 and 2 — never reaches it. Only genuinely ambiguous cases land here, keeping the total system latency below 1 millisecond.

2.3 The Final Decision Rule

The three layers are combined with a logical OR rule: if ANY layer says 'attack', the final answer is 'attack'. Think of three separate alarm systems in a building. If the smoke detector, heat sensor, OR motion sensor goes off — the fire department is called. You do not need all three to agree. This maximises recall: no attack slips through.

3. Methodology

How the Research Was Actually Conducted, Step by Step

3.1 Where the Data Came From

The Canadian Institute for Cybersecurity built a controlled lab network, ran real DDoS attacks against it, and recorded every packet — both the attack traffic and the normal traffic. This produced 283,031 labelled examples, each tagged as either 'legitimate user' or 'attack packet.' This dataset is called CIC-DDoS2019 and is the standard worldwide benchmark for comparing DDoS detection systems.

Why a Lab Dataset and Not Real Internet Traffic?

Privacy and legality: Capturing real people's internet traffic without consent is illegal and raises serious ethical issues.

Ground truth: A lab dataset provides exact labels — researchers know with certainty which packets were attacks and which were legitimate.

Reproducibility: Any other researcher can download the same dataset and repeat the experiment, enabling fair comparisons.

3.2 Training the Four AI Models

80% of the 283,031 examples were used to teach the AI models (training set). The remaining 20% — examples the AI had never seen — were used to test how well it learned (test set). This is exactly how a student studies from textbooks and then takes an exam on new questions they have not seen before.

AI Model	How It Thinks — The Intuition
Random Forest	Builds 100 different decision trees, each trained on a slightly different random sample of the data. All 100 trees vote; the majority wins. Like asking 100 doctors for a diagnosis and going with the consensus.
XGBoost	Builds trees one after another, each one specifically learning from the mistakes of the previous one. Like a student who reviews only the questions they got wrong in each practice test, getting progressively better.
GRU (Neural Network)	Looks at sequences of traffic windows over time, remembering what happened in previous seconds. Like a doctor who considers a patient's full medical history, not just today's symptoms. Very accurate — but too slow for real-time deployment.
Isolation Forest	Trained only on normal traffic. Does not need to know what attacks look like — it only knows 'normal'. Anything statistically far from normal gets flagged. Like a doctor who has only ever seen healthy patients and immediately recognises when something is wrong.

Table 2: The four AI models and their intuitive working principles.

4. Results

What Do the Numbers Actually Mean for Real People?

4.1 The Metrics Explained in Plain English

Metric	What It Means in Plain English
Accuracy	Out of all traffic examined (attacks and normal combined), how many were correctly classified? 99.99% means out of 10,000 cases, only 1 was wrong.
Recall	Out of all the actual attacks, how many did the system correctly identify as attacks? 99.99% means out of 10,000 real attacks, only 1 was missed. This is the most critical number in security — a missed attack is catastrophic.
F1 Score	A combined score from 0 to 1 balancing both precision and recall. 1.000 is perfect — no attacks missed, no legitimate traffic wrongly blocked.
Latency (Speed)	How long does detection take? In network security, everything must happen in under 1 millisecond — faster than a camera flash — or the attack is already causing damage.

Table 3: Performance metrics explained for a non-technical reader.

4.2 The Performance Results

Model	Accuracy	Recall	F1 Score	Speed (ms)
Random Forest	99.98%	99.98%	0.9998	0.41
XGBoost	99.97%	99.96%	0.9997	0.33
GRU (Neural Network)	99.60%	99.50%	0.9950	2.80 
Isolation Forest	97.00%	96.00%	0.9600	0.18
Hybrid (All 3 Layers)	99.99%	99.99%	1.000	0.62

Table 4: Model comparison. GRU's 2.80ms latency  exceeds the 1ms real-time limit.

4.3 What Do These Numbers Mean for Real People?

BEFORE — Without This System	AFTER — With This System
A bank website goes down. Customers cannot check balances, make payments, or transfer money for hours.	Attack detected in under 1 millisecond — before any user notices a slowdown. The bank stays fully online.

A hospital patient record system is taken offline. Doctors cannot access medication history during emergencies.	The attack is flagged at Layer 1 in microseconds. Hospital systems remain continuously available for patient care.
An e-commerce site crashes on a sale day. Every minute of downtime costs █80,000+ in lost sales revenue.	The hybrid system stops the attack before checkout is impacted. Sales continue uninterrupted.
Security analysts receive thousands of alerts per day. Over 90% are false alarms. Real attacks get buried.	99.99% accuracy means almost all alerts are genuine. Analysts focus on real threats — not noise.
A brand-new attack type (zero-day) bypasses all signature-based defences — it has never been seen before.	Isolation Forest (Layer 2) detects it as statistically abnormal, even without knowing what type of attack it is.

Table 5: Real-world impact before and after this detection system.

4.4 Only 4 Attacks Missed Out of 45,234

The confusion matrix shows exactly how many cases the system got right and wrong across 56,606 test examples:

	System says: SAFE	System says: ATTACK
Actually: SAFE	11,328 ✓ Correctly identified safe	44 ✗ Safe traffic wrongly flagged (0.39%)
Actually: ATTACK	4 ✗ Attacks that slipped through	45,230 ✓ Attacks correctly detected

Table 6: Confusion matrix — hybrid model on 56,606 test cases. Only 4 real attacks were missed.

In plain terms: out of 45,234 real attacks, only 4 slipped through — an error rate of 0.009%. The 44 false alarms (safe traffic wrongly flagged) equal roughly 15 misidentified flows per minute in a production environment — easily manageable with a brief manual review.

5. Future Projects

Why This Research Is the Base for What Comes Next

5.1 What This Research Establishes as a Foundation

Foundation	How Future Projects Benefit
Benchmark Numbers	Anyone building a new DDoS system now has a clear target: beat 99.99% accuracy, F1 = 1.000, latency < 1ms on CIC-DDoS2019. Without a baseline, progress cannot be measured or compared.
Sliding Window Method	The five-feature, one-second window with 50% overlap is a documented, tested technique for real-time traffic analysis. Future projects can use it directly without reinventing it.
Three-Layer Architecture	The cascade of Rule → Anomaly → Supervised is a proven design pattern for real-time detection. Any detection system — not just DDoS — can adopt this: cheap fast checks first, expensive thorough checks last.
Four-Model Comparison	First paper to benchmark all four model families under identical conditions. Future researchers can cite this work and immediately begin extending it, rather than repeating the same baseline experiments.

Table 7: What this research establishes as a reusable starting point.

5.2 Six Direct Extensions — Where the Next Project Begins

EXTENSION 1 — Adaptive Thresholds

The current Layer 1 thresholds (50,000 packets/sec, 10 bytes² variance) were manually tuned for one dataset. A future project could use online machine learning to automatically adjust these thresholds as network conditions change — making the system work on any network without manual recalibration.

Where to start: Use an online gradient descent algorithm that updates the thresholds every 60 seconds based on the last hour of observed traffic patterns.

EXTENSION 2 — Federated Learning (Privacy-Preserving)

All training data is currently gathered in one place — a privacy problem. Internet providers cannot share customer traffic data. Federated learning trains models across many organisations without any raw data leaving each one.

Where to start: Simulate 5 virtual ISPs each with local data. Train a shared model without centralising any data using the Flower or PySyft framework.

EXTENSION 3 — Encrypted Traffic Detection

As TLS encryption becomes universal, packet payload features become unreadable. The current system uses only header-level statistics which remain visible. Future work measures whether the same accuracy holds with metadata alone.

Where to start: Filter CIC-DDoS2019 to only header-level features (packet count, timing, size). Measure the accuracy drop — the size of that gap is the research contribution.

EXTENSION 4 — Multi-Class Attack Classification

This paper uses binary classification: attack or not attack. A more useful system identifies the specific attack type (UDP Flood, SYN Flood, DNS Amplification) so defenders know exactly what countermeasure to apply.

Where to start: Extend the Random Forest from binary to 8-class classification using the 8 attack categories already labelled in CIC-DDoS2019.

EXTENSION 5 — IoT Device Deployment

IoT devices (smart cameras, home routers) have very limited processing power. A 100-tree Random Forest cannot run on a Raspberry Pi. Future work compresses the model to run on tiny embedded hardware.

Where to start: Train a large RF model, then use model distillation to produce a tiny version. Deploy on a Raspberry Pi 4 and measure the accuracy versus latency trade-off.

EXTENSION 6 — Adversarial Robustness Testing

An attacker who knows the detection pipeline can craft traffic to slip past it. Future projects can generate adversarial DDoS samples that fool the system, then use those samples to retrain the system to become harder to fool.

Where to start: Use the cleverhans or ART (Adversarial Robustness Toolbox) library to generate adversarial DDoS samples. Measure accuracy drop, then retrain using those adversarial examples.

5.3 The Three-Layer Pattern Applies Everywhere

The Rule → Anomaly Detector → Supervised Classifier cascade is a general design pattern for any real-time classification problem. The same structure appears in:

- Email spam filtering — Rule layer blocks known spam domains. Anomaly layer flags unusual senders. Supervised layer classifies ambiguous content.
- Credit card fraud — Rule layer blocks impossible transactions (card used in two countries simultaneously). Anomaly catches unusual spending. Supervised layer decides borderline cases.
- Medical imaging triage — Rule layer screens clearly healthy scans instantly. Anomaly layer flags anything unusual. Supervised layer provides specific diagnoses.
- Industrial monitoring — Rule layer catches sensor readings outside safe limits. Anomaly layer detects developing faults. Supervised layer identifies the specific failure type.

Understanding this paper means understanding a design pattern that appears throughout modern AI systems. That is exactly why it makes an excellent foundation for future work in any real-time classification domain.

6. Summary

The Complete Picture in One Place

The Whole Paper in One Paragraph

DDoS attacks flood websites with so much fake traffic that real users cannot get through. Old detection systems were either too slow (30+ seconds), too rigid (only catching known attacks), or too inaccurate. This research built a three-layer system: Layer 1 applies simple rules and catches obvious floods in under 0.01 milliseconds; Layer 2 uses an AI trained only on normal traffic to detect unusual behaviour — including brand-new attack types never seen before; Layer 3 uses a 100-tree voting AI for precise classification of ambiguous cases. Together, the three layers achieved 99.99% accuracy, F1 = 1.000, and detected attacks in under 1 millisecond — faster than a single camera flash — missing only 4 attacks out of 45,234. Most importantly, the three-layer cascade architecture is a reusable design pattern for any real-time detection problem.

The Research Journey at a Glance

Stage	Key Point
Problem	Millions of fake packets per second flood a server. Old detection takes 30–120 seconds. The damage is already done.
Dataset	CIC-DDoS2019 — 283,031 labelled examples of real attacks and legitimate traffic from a controlled lab environment.
Feature Extraction	Five numbers computed from every one-second traffic window: packet rate, mean size, size variance, inter-arrival time, byte rate.
Layer 1: Rules	Packet rate > 50,000/sec OR size variance < 10 bytes ² → instant attack flag. Speed: 0.01ms. Catches volumetric floods.
Layer 2: iForest	Trained on normal traffic only. Any traffic too different from normal gets flagged. Catches zero-day attacks. Speed: 0.18ms.
Layer 3: RF	100 decision trees, majority vote, full 68 features. Expert-level decision for ambiguous traffic. Speed: 0.41ms.
Combined Result	99.99% accuracy · F1 = 1.000 · Only 4 attacks missed out of 45,234 · Total latency 0.62ms.
Real Impact	Banks stay online. Hospitals stay accessible. E-commerce keeps selling. Security analysts deal with far fewer false alarms.
Future Work	6 clear extensions: adaptive thresholds, federated learning, encrypted traffic, multi-class detection, IoT deployment, adversarial hardening.

Table 8: The complete research journey from problem to future directions.

The Core Lesson

When a problem is too big and fast for any single tool, build layers. Make each layer fast and cheap. Make each layer cover what the previous one missed. The whole will always be greater than any of its individual parts.

This is not just a lesson about DDoS. It is a way of thinking about any hard real-time detection problem — in security, medicine, finance, or industry.